

Modeling Computation

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Contents

Preface	3
1 Alphabets, Languages, and Grammars	1
1.1 Terminology	1
1.2 (Phrase-Structure) Grammars	1
2 Types of Grammars	3
2.1 Backus-Naur Form (BNF) for Context Free Grammars	4
3 Finite State Automata	5
3.1 Graph Representation	5
3.2 Extended transition function	5
3.3 Language Recognition	6
3.4 Equivalent Automata and Grammars	7
3.5 Nondeterministic Automata	7
3.6 The Equivalence of DFA and NFA.	9
4 Regular Expressions and Languages	10
4.1 Operations on Languages	10
4.2 Regular Expressions	11
4.3 Recognizing Regular Languages with NFAs	12
5 Regular Grammars and Automata	13
5.1 Grammars to Automata	13
5.2 Automata to Grammars	14
5.3 Some Review	15
6 Regular Languages	16
6.1 Equivalent Representations	16
6.2 DFA to Regular Expression	17
6.3 Review problems	19
7 A language that is not regular	20
8 Turing Machines	21
8.1 Definition of a 1-tape Turing Machine (Syntax)	21
8.2 Execution of a Turing Machine program (Semantics)	21
8.3 ASCII Representations of Turing Machines	22
8.4 The language recognized by a Turing Machine	24
8.5 Computing functions with a Turing machine	24

9	What Can Turing Machines Do?	25
9.1	Turing Computable Languages	25
9.2	Variations on the Turing Machine	25
9.3	The Halting Problem	26
9.4	Running time of a Turing machine	28

Preface

These exercises were assembled to accompany a Discrete Mathematics course at Calvin College.

1 Alphabets, Languages, and Grammars

1.1 Terminology

- An **alphabet** (or **vocabulary**) is just a finite, non-empty set. Its elements are called **letters** or **symbols**.
- A **word** (or **string**) is a finite sequence of symbols.
- A word consisting of no symbols is called the **empty word** and denoted λ . (Think ".")
- The set of all words using exactly n symbols from V (repetition allowed) is denoted V^n .
- The set of all words using symbols in alphabet V is denoted V^* . (So $V^* = V^0 \cup V^1 \cup V^2 \dots$)
- A **language** over V is a subset of V^* .

Exercises

1. Let $V = \{0, 1\}$. What is V^0 ? What is V^2 ? What is V^* ?
2. Let $V = \{2\}$. What is V^0 ? What is V^2 ? What is V^* ?
3. Can $\emptyset$ (empty set) be a letter? an alphabet? a word? a language?

1.2 (Phrase-Structure) Grammars

A phase-structure grammar consists of

- an **alphabet**: denoted V here
- a **start symbol** S : S must be (exactly) one of the symbols in the alphabet. We can denote this as $S \in V$.
- **terminal symbols** (T): a set of symbols in the alphabet ($T \subset V$)
 - $N = V - T$ is the set of **nonterminal symbols**
 - so every symbol is either terminal or nonterminal
 - The start symbol is a terminal in some languages and a nonterminal in others
- **production rules** (P): $P \subseteq (V^* - T^*) \times V^*$
 - usually we write elements of P as $u \rightarrow v$ rather than (u, v) .
 - u : $V^* - T^*$ says that the lefthand side of the rule (u) must **contain at least one nonterminal**.
 - v : The righthand side can be any combination of terminals and nonterminals (including the empty string).

Examples

Note: in each of these examples, capital letters are used for nonterminals and lower case letters or digits for terminals. That makes it easy to remember, but it is not a requirement of the definition of a grammar.

Grammar 1 (G_1): alphabet: $\{a, b, A, B, S\}$, terminals: $\{a, b\}$, start symbol: S , production rules:

- $S \rightarrow ABa$
- $A \rightarrow BB$
- $B \rightarrow ab$
- $AB \rightarrow b$

Grammar 2 (G_2): alphabet: $\{S, A, a, b\}$, terminals: $\{a, b\}$, start symbol: S , production rules:

- $S \rightarrow aA$
- $S \rightarrow b$
- $A \rightarrow aa$

Grammar 3 (G_3): alphabet: $\{S, 0, 1\}$, terminals: $\{0, 1\}$, start symbol: S , production rules:

- $S \rightarrow 11S$
- $S \rightarrow 0$

Grammar 4 (G_4): alphabet: $\{A, B, C, D, a, b, c\}$, terminals: $\{a, b, c\}$, start symbol: A , production rules:

- $A \rightarrow BC$
- $B \rightarrow Da$
- $C \rightarrow Ca$
- $C \rightarrow Db$
- $C \rightarrow b$
- $D \rightarrow cb$
- $D \rightarrow b$

1.2.1 Derivations and Languages

The rules of a grammar are used to derive strings of terminals (elements of T^*) as follows.

- If $w_0 \rightarrow w_1$ is a rule, then $lw_0r \Rightarrow lw_1r$ for any strings l and r .
- $a \xRightarrow{*} b$ is defined recursively. $a \xRightarrow{*} b$ if either
 - $a \Rightarrow b$, or
 - there is a c such that $a \Rightarrow c \wedge c \xRightarrow{*} b$

Basically the production rules are “rewrite rules” that allow us to replace the left side of the rule with the right side. A derivation is a sequence of rewrites.

The language of a grammar G is denoted $L(G)$ and contains all strings of terminals that can be derived from the start symbol:

$$L(G) = \{w \in T^* \mid S \xRightarrow{*} w\}$$

Exercises

4. Show that using Grammar 1, $ABa \xRightarrow{*} abababa$. Show every step in the process.
5. Is $abababa \in L(G_1)$, the language generated by Grammar 1?
6. What is $L(G_2)$?
7. What is $L(G_3)$?
8. Generate several words using G_4 .
9. Create a grammar G_5 such that $L(G_5) = \{0^n 1^n \mid n = 0, 1, 2, \dots\}$
10. Create a grammar G_6 such that $L(G_6) = \{0^n 1^n \mid n \in \mathbb{Z}^+\}$
11. Create a grammar G_7 such that $L(G_7) = \{0^m 1^n \mid m, n \in \mathbb{N}\}$

2 Types of Grammars

Type 0: no restrictions

Type 1 (context sensitive): only two types of rules allowed

- $lAr \rightarrow lwr$ where
 - $l, r \in V^*$ [so they can be anything, including empty]
 - $A \in N$ [so A is a single non-terminal]
 - $w \in V^+$ [That is, $w \neq \lambda$.]
 - shorthand: $[\text{left}] [\text{nonterminal}] [\text{right}] \rightarrow [\text{left}] [\text{non-empty}] [\text{right}]$
- $S \rightarrow \lambda$
 - shorthand: $[\text{start}] \rightarrow [\text{empty string}]$
 - **If this rule is used, the the start symbol S may not appear on the right side of any rule.**

Type 2 (context free): only one type of rule allowed

- $A \rightarrow w$ where
 - $A \in N$ [so A is a single nonterminal symbol]
 - $w \in V^*$ [so no restriction here]
- shorthand: $[\text{nonterminal}] \rightarrow [\text{any string}]$

Type 3 (regular): Three types of rules allowed

- $[\text{nonterminal}] \rightarrow [\text{terminal}]$ [non terminal] [Example: $A \rightarrow bC$]
- $[\text{nonterminal}] \rightarrow [\text{terminal}]$ [Example: $A \rightarrow b$]
- $[\text{start}] \rightarrow [\text{empty string}]$ [Example: $S \rightarrow \lambda$]

A regular/context free/context sensitive language is a language that can be generated by a regular/context free/context sensitive grammar.

1. For Grammars 1 - 4 from the previous section, determine whether it is regular, context free, or context sensitive. (Some grammars will be more than one. All grammars are type 0.)
2. Consider **Grammar 8** (G_8): alphabet: $\{A, B, C, x, y, z\}$, terminals: $\{a, b, c\}$, start symbol: A , production rules:
 - $A \rightarrow \lambda$
 - $A \rightarrow xBC$
 - $BC \rightarrow xyC$
 - $B \rightarrow x$
 - $B \rightarrow AC$
 - $xCx \rightarrow zyz$
 - $C \rightarrow Ax$
 - $C \rightarrow yB$
 - $C \rightarrow aBc$
 - $C \rightarrow z$
 - a. Why isn't this a type 1 grammar?
 - b. Which rules can you eliminate to make this a type 1 grammar? (Eliminate as few as possible.)
 - c. Which rules can you eliminate to make this a type 2 grammar? (Eliminate as few as possible.)
 - d. Which rules can you eliminate to make this a type 3 grammar? (Eliminate as few as possible.)
3. True or false: Every grammar of type n is a grammar of type $n - 1$.
4. True or false: Every language of type n is a language of type $n - 1$.

2.1 Backus-Naur Form (BNF) for Context Free Grammars

Shorthand notation for type 2 grammars.

- nonterminals denoted with $\langle \rangle$.
- $\rightarrow$ written as $::=$
- All rules with same nonterminal on left written together with the list of possible right sides separated by $|$.

Example: If we have production rules $A \rightarrow Aa$, $A \rightarrow a$, and $A \rightarrow AB$, we will write

$\langle A \rangle ::= \langle A \rangle a \mid a \mid \langle A \rangle \langle B \rangle$

BNF for ALGOL 60 identifier

$\langle \text{identifier} \rangle ::= \langle \text{letter} \rangle \mid \langle \text{identifier} \rangle \langle \text{letter} \rangle \mid \langle \text{identifier} \rangle \langle \text{digit} \rangle$
 $\langle \text{letter} \rangle ::= a|b|c|d|e|f|g|h|i|j|k|l|m|n|o|p|q|r|s|t|u|v|w|x|y|z$
 $\langle \text{digit} \rangle ::= 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

BNF for signed integer

$\langle \text{signed integer} \rangle ::= \langle \text{sign} \rangle \langle \text{integer} \rangle$
 $\langle \text{sign} \rangle ::= + \mid -$
 $\langle \text{integer} \rangle ::= \langle \text{digit} \rangle \mid \langle \text{digit} \rangle \langle \text{integer} \rangle$
 $\langle \text{digit} \rangle ::= 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

BNF specification for Java

You can find BNF for various programming languages online. For example: <https://users-cs.au.dk/amoeller/RegAut/JavaBNF.html> has a BNF specification for Java.

Exercises

5. Why does BNF notation only work for context free grammars?
6. For each context free grammar we have seen, give its BNF representation.

3 Finite State Automata

A **finite state automaton** consists of the following:

- a finite set of **states**: S
- a finite **input alphabet**: I
- a **start state**: $s_0 \in S$.
- a set of **accepting states**: $F \subseteq S$
- a **transition function**: $f : S \times I \rightarrow S$

Example (M_0) states: $\{A, B, C, D\}$; input alphabet: $\{0, 1\}$; start state: A ; accepting states: $\{A, D\}$; transition function described in table below

M_0								
state	A	A	B	B	C	C	D	D
letter	0	1	0	1	0	1	0	1
f	A	B	A	C	A	A	C	B

3.1 Graph Representation

1. It is often easier to visualize what is going on if we represent an automaton with a graph. Create a labeled, directed graph with the following properties:
 - There is a vertex for each state, labeled with the state. (Draw this as a circle with the state inside the circle.)
 - There is an edge from state s to state t labeled with letter x if and only if $f(s, x) = t$
 - Accepting states are circled a second time. (We could use shape or color or something else to make it clear which states are accepting states, but double circling is easy.)
 - An extra arrow (not coming from any state) points to the start state. This isn't really an edge in the graph, just an extra bit of labeling.

3.2 Extended transition function

We can extend the transition function to $f^* : S \times I^* \rightarrow S$ with the following recursive definition:

- $f^*(s, \lambda) = s$ for any state s
 - $f^*(s, xa) = f(f^*(s, x), a)$ for any $x \in I^*$ and $a \in I$
2. Use this definition to determine the following.
 - a. $f^*(B, \lambda)$
 - b. $f^*(B, 0)$
 - c. $f^*(B, 010)$
 - d. $f^*(A, 101)$
 - e. $f^*(A, 1011)$

3.3 Language Recognition

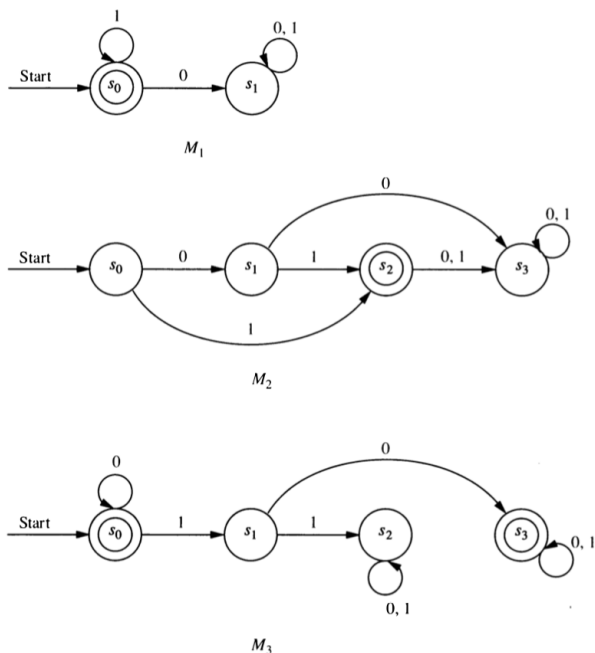
For any automaton M with transition function f , start state s and accepting states F , the language recognized by M (written $L(M)$) is defined as follows:

$$x \in L(M) \iff f^*(s, x) \in F$$

3. For each string x below, compute $f^*(A, x)$. Which of these strings are in $L(M_0)$? (We will say that such strings are **accepted by** M_0 .)

- a. λ
- b. 0
- c. 1
- d. 010
- e. 1011

Here are three automata:

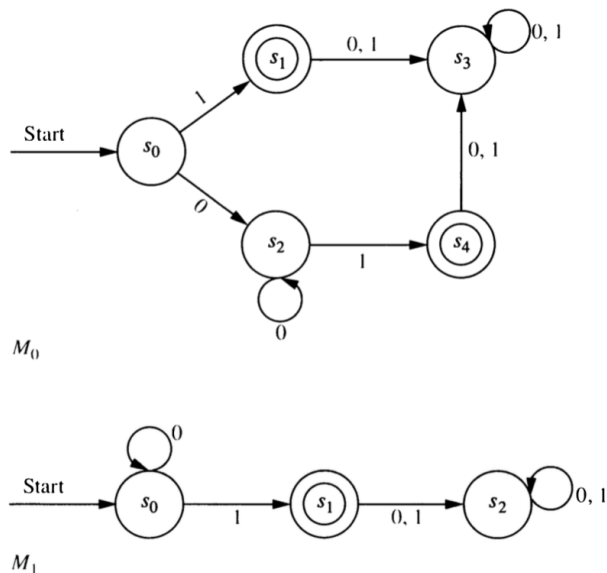


4. For each of the machines M_1, M_2, M_3 , compute
- a. $f^*(s_0, 10)$
 - b. $f^*(s_0, 1011)$
 - c. $f^*(s_1, 1011)$
5. For each of the machines M_1, M_2, M_3 , determine the language recognized.
6. Create automata that recognize each of the following languages.
- a. The set of bitstrings that begin with two 0's
 - b. The set of bitstrings that contain exactly two 0's
 - c. The set of bitstrings that contain at least two 0's
 - d. The set of bitstrings that contain two consecutive 0's (anywhere in the string)
 - e. The set of bitstrings that do not contain two consecutive 0's anywhere
 - f. The set of bitstrings that end with two 0's

3.4 Equivalent Automata and Grammars

We will say that two automata M and N are equivalent if $L(M) = L(N)$, that is, if they recognize the same language. Similarly, two grammars are equivalent if they generate the same language.

7. What must you do to show that two automata are **not** equivalent?
8. What must you do to show that two automata **are** equivalent.
9. Are the two automata represented below equivalent? Explain.



3.5 Nondeterministic Automata

The finite state automata we have seen so far are often called **deterministic finite-state automata** or DFAs. There is a generalization called a **non-deterministic finite-state automaton** or NFA. The only difference is how the transition function is specified. For an NFA, the transition function has the form

$$f : S \times I \rightarrow P(S)$$

where S is the set of states, I is the input alphabet, and $P(S)$ is the powerset of S . (The powerset of S is the set of all subsets of S .) So the transition function for an NFA returns a *set* of states. That set could contain just one state, or multiple states, or no states at all (empty set).

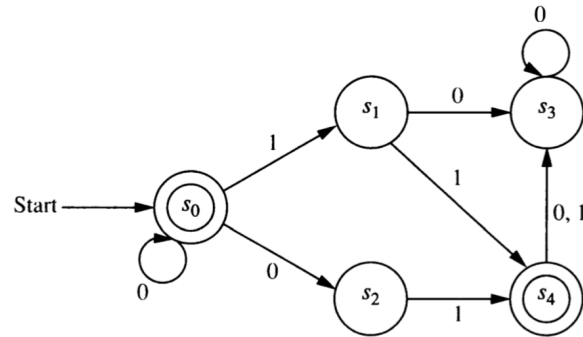
Here is a tabular description of the transition function for an NFA N_0 with start state A and accepting states C and D.

N_0								
state	A	A	B	B	C	C	D	D
letter	0	1	0	1	0	1	0	1
f	$\{A, B\}$	$\{D\}$	$\{A\}$	$\{B, D\}$	$\{\}$	$\{A, C\}$	$\{A, B, C\}$	$\{B\}$

10. Draw the graph representation. How will the graph of an NFA differ from the graph for a DFA?

11. How do we define the language recognized by an NFA? [Check your answer before proceeding.]
12. Which of these strings are accepted by N_0 ?
 - a. 01
 - b. 011
 - c. 00
 - d. 0000
 - e. λ
 - f. 010101

Here is the graph of an NFA N_1 .



13. Make a table showing the transition function for N_1 .
14. Which of the following strings are accepted by N_1 ?
 - a. λ
 - b. 00
 - c. 01
 - d. 010
 - e. 011
 - f. 001
15. Which is harder: to convince someone that a string *is* accepted by an NFA or that it *is not* accepted by an NFA? Why?
16. What is $L(N_1)$?
17. For each of the following languages
 - Construct an NFA that recognizes the language.
 - Construct a DFA that recognizes the language.

In each case, try to do this with as few states and transitionas as you can. You may find it helpful to give your states good names that reflect what they represent.

 - a. The set of bitstrings that either start 01 or start 10.
 - b. The set of bitstrings that end with two 0's
 - c. The set of bitstrings that either end 01 or end 10.

Which do you find easier to construct: NFAs or DFAs? Why?
18. Create a DFA that recognizes $L(N_1)$.
19. Given an NFA N , is it always possible to create a DFA M that recognizes the same language? If so, explain how. If not, provide an example N and explain why $L(N)$ cannot be recognized by a DFA.

3.6 The Equivalence of DFA and NFA.

Despite their apparent differences, DFAs and NFAs recognize exactly the same set of languages. That is

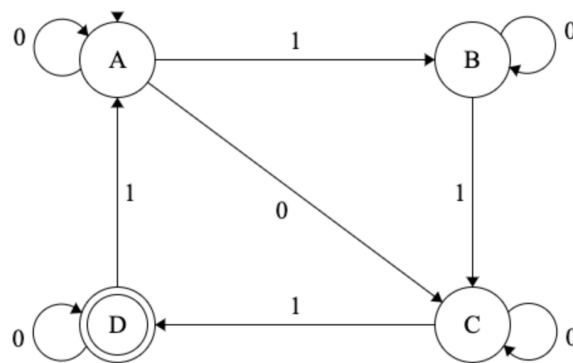
- If the language L is recognized by a DFA M , then there is also an NFA N that recognizes L .
- If the language L is recognized by an NFA N , then there is also a DFA M that recognizes L .

To prove this, we need

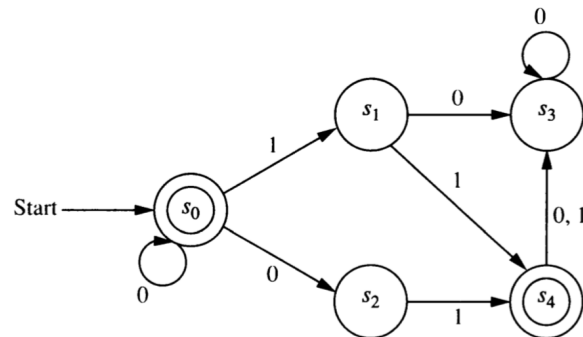
- Algorithm 1: an algorithm that converts DFAs into equivalent NFAs
- Algorithm 2: an algorithm that converts NFAs into equivalent DFAs.

20. One of these two algorithms is super easy. Which one? What is the algorithm?

21. The other direction is more interesting. Let's see if we can figure out a general algorithm for this task by first exploring an example. Consider the following NFA. Create an equivalent DFA, using a method that could be applied to any NFA.



22. Now convert the NFA N_1 into an equivalent DFA. Here's N_1 again.



23. If an NFA has n states and we use this method to convert it to a DFA, what is the most states the DFA might have?

4 Regular Expressions and Languages

4.1 Operations on Languages

Since languages are sets, we can use set operations like $\cup$ and $\cap$ to create new languages. But there are some additional operations that we will use.

- concatenation of strings: xy is the concatenation of strings x and y (x followed by y)
- concatenation of sets: $AB = \{xy \mid x \in A \wedge y \in B\}$
- power: $A^0 = \emptyset$; $A^1 = A$; $A^{n+1} = A^n A$. (Power is iterated concatenation.)
- Kleene star: $A^* = \cup_{n=0}^{\infty} A^n$

1. What strings belong to $\{0, 01\}^2$?
2. Which of these strings belong to $\{0, 01\}^*$?
 - a. 01001
 - b. 10110
 - c. 00010
3. Which of these strings belong to $\{101, 111, 11\}^*$?
 - a. 1010111
 - b. 1011011
 - c. 1110111
 - d. 11110111

4.2 Regular Expressions

Regular expression are a type of shorthand notation for representing certain types of sets.

4.2.1 Syntax

A regular expression over an input alphabet I is defined recursively.

- $\emptyset$ and λ and regular expressions.
- x is a regular expression for each $x \in I$.
- If A and B are regular expressions, then
 - A^* is a regular expression
 - $(A \cup B)$ is a regular expression
 - (AB) is a regular expression

4.2.2 Semantics

Each regular expression represents a language as follows.

- $\emptyset$ represents the empty language: $\emptyset$
- λ represents language that contains only the empty string: $\{\lambda\}$
- For each $x \in I$, x represents the language containing only x : $\{x\}$
- If A and B are regular expressions representing languages A and B , then
 - (AB) represents AB
 - $(A \cup B)$ represents $A \cup B$ [Note: This is often written ‘ $A|B$ ’ in programming languages.]
 - A^* represents A^* [Note: This is often written as ‘ A^* ’ in programming languages.]

The languages represented by regular expressions are called **regular languages** (how creative).

4. In Rosen 7, the definition of regular expression is missing boldface in a couple places. Find them.

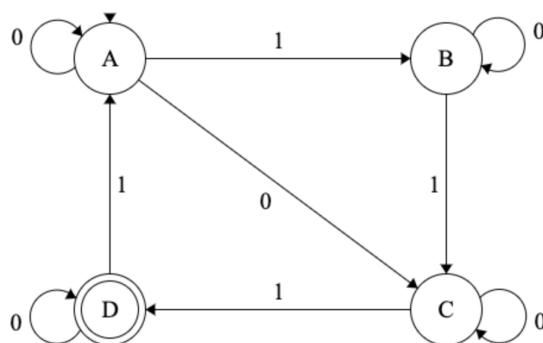
The *regular expressions* over a set I are defined recursively by:

the symbol $\emptyset$ is a regular expression;
the symbol λ is a regular expression;
the symbol x is a regular expression whenever $x \in I$;
the symbols (AB) , $(A \cup B)$, and A^* are regular expressions whenever A
and B are regular expressions.

5. What is the difference between **1** and 1? between **λ** and λ ? between **$\emptyset$** and $\emptyset$?
6. Describe in words the languages represented by each of these regular expressions:
a. **10^*** b. **$(10)^*$** c. **$0 \cup 01$** d. **$0^* \cup 01$** e. **$0(0 \cup 1)^*$** f. **$(0 \cup 01)^*$**
7. Can each of the languages above be recognized by an NFA? a DFA?
8. What would we need to do to show that every regular language can be recognized by an NFA? Give it a try and see how far you get. Which parts are easy? Which are trickier? Why?

4.3 Recognizing Regular Languages with NFAs

1. Show that the empty language ($\emptyset$) is recognized by an NFA.
2. Show that the language $\{\lambda\}$ is recognized by an NFA.
3. Show that if $a \in I$, then the language $\{a\}$ is recognized by an NFA.
4. An NFA is said to have a **lonely start state** if there are no transitions into the start state.
 - a. Construct an NFA with a lonely start state that is equivalent to the NFA below.



- b. Explain why every NFA is equivalent to an NFA with a lonely start state. That is, describe how to convert any NFA into an equivalent NFA with a lonely start state.
5. Show that if A and B are each recognized by an NFA, then AB is, too.

[Hint: Let N_A and N_B be the NFAs that recognize A and B . You may assume each has a lonely start state if that is useful. How can you use them to build a new NFA that recognizes AB ?]
6. Show that if A and B are each recognized by an NFA, then $A \cup B$ is, too.

[Hint: Let N_A and N_B be the NFAs that recognize A and B . You may assume each has a lonely start state if that is useful. How can you use them to build a new NFA that recognizes $A \cup B$?]
7. Show that if A is recognized by an NFA, then A^* is, too.

[Hint: Let N_A be the NFA that recognizes A . You may assume it has a lonely start state if that is useful. How can you use it to build a new NFA that recognizes A^* ?]
8. Explain how 1–7 above show that every regular language is recognized by an NFA.
9. Explain how 7 implies that every regular language is recognized by a DFA.
10. The method just outlined is automatic (you could fairly easily write a computer program to do the translation from regular expression to NFA), and it provides a proof that all regular languages can be recognized by an NFA. But it might produce an NFA with more states than is minimally required. Create an NFA that recognizes $1^* \cup 01$. Do this two ways:
 - a. Following the steps outlined in 1–6.
 - b. Any way you like, but using fewer states than in part a.
 - c. What is the smallest (fewest states) NFA you can find that recognizes $1^* \cup 01$?
11. Show that if A is recognized by an NFA, then the complement of A ($\bar{A} = A^c$) is too.

[Hint: There is an easy way and a hard way – use the easy way.]

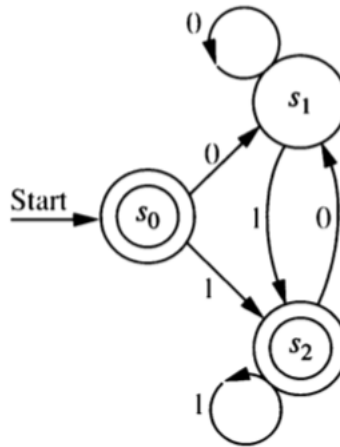
5 Regular Grammars and Automata

5.1 Grammars to Automata

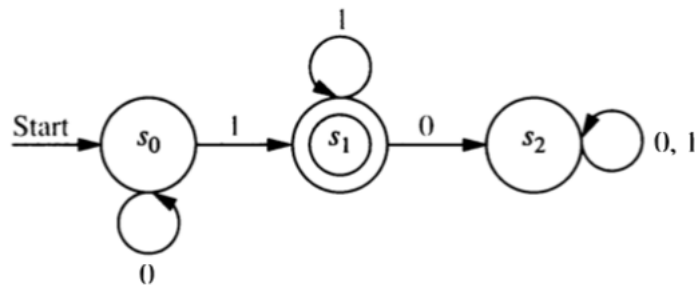
1. Construct a finite state automaton that recognizes the language generated by the following regular grammar.
 - terminals: 0 and 1
 - nonterminals: S and A
 - start symbol: S .
 - production rules: $S \rightarrow 1A$, $S \rightarrow 0$, $S \rightarrow \lambda$, $A \rightarrow 0A$, $A \rightarrow 1A$, $A \rightarrow 1$
2. Construct a finite state automaton that recognizes the language generated by the following regular grammar.
 - terminals: 0 and 1
 - nonterminals: S , A , B
 - start symbol: S .
 - production rules: $S \rightarrow 1A$, $S \rightarrow 0$, $A \rightarrow 0B$, $A \rightarrow 1A$, $A \rightarrow 1 B \rightarrow 0A$, $B \rightarrow 1B$, $B \rightarrow 0$
3. Explain how *any* regular grammar G can be converted into an NFA N such that $L(N) = L(G)$.
 - a. What is a regular grammar? (What sorts of production rules are allowed?)
 - b. What will the states of N be? How many will there be?
 - c. What state will be the start state?
 - d. Which states will be final (ie. accepting) states?
 - e. How does each rule in G get converted into a part of N ? [Go through each kind of rule a regular grammar may have.]

5.2 Automata to Grammars

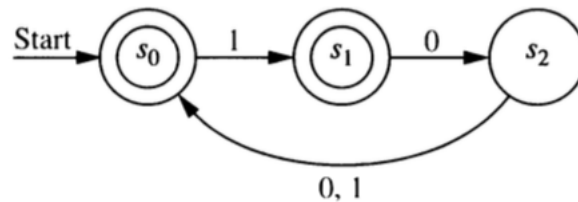
4. Show that every DFA or NFA is equivalent to a DFA or NFA with a “lonely start state.” A lonely start state is a start state that can never be returned to. (In terms of the graph, its in-degree is 0.)
5. Convert each of the automata in the next three problems into an automaton with a lonely start state.
6. Construct a regular grammar that generates the language accepted by this finite state automaton.



7. Construct a regular grammar that generates the language accepted by this finite state automaton.



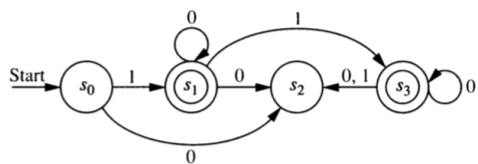
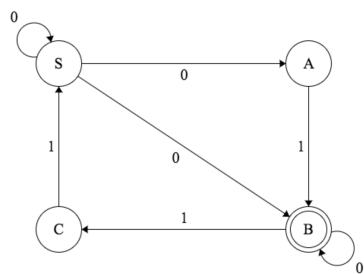
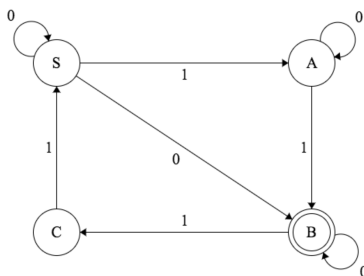
8. Construct a regular grammar that generates the language accepted by this finite state automaton.



9. In this problem you will show that any DFA or NFA with a lonely start state is equivalent to a regular grammar by showing how to construct a grammar G from such an automaton.
 - a. What will the terminals and nonterminals in your grammar be?
 - b. What will the start symbol be?
 - c. How are the production rules created?
 - d. Why was it necessary to have a lonely start state?

5.3 Some Review

1. Convert each of the following NFA's into an equivalent DFA. (Remember: equivalent means that they recognize the same language.) Do this using our general algorithm for this task. For the first two, the start state is S .



6 Regular Languages

6.1 Equivalent Representations

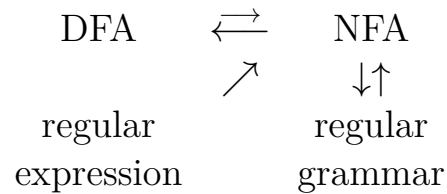
We have been working toward a result that says the following are all equivalent.

1. L can be described by a regular expression.
2. L is recognized by a DFA.
3. L is recognized by an NFA.
4. L is generated by a regular grammar.

The proof that these are equivalent amounts to describing algorithms that can convert from one representation to another. So far we have shown

- a. $1 \Rightarrow 3$
- b. $2 \Rightarrow 3$
- c. $3 \Rightarrow 2$
- d. $2 \Rightarrow 4$
- e. $4 \Rightarrow 3$

We can depict this a bit more visually like this:



Still missing from our list is

- f. $? \Rightarrow 1$

It doesn't matter which representation we start from (do you see why?), so we will choose the easiest:

DFA $\Rightarrow$ regular expression

Once we have that, we will see that all 4 definitions are equivalent.

6.2 DFA to Regular Expression

6.2.1 GNFA (Generalied NFA)

A GNFA (Generalized NFA) is like an NFA but the edges/transitions may be labeled with **any regular expression**. One way of obtaining a regular expression from a DFA uses an algorithm that works with GNFA's.

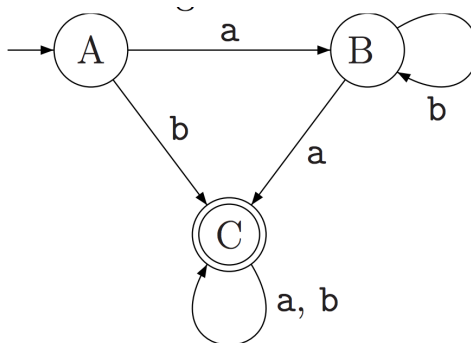
6.2.2 Algorithm for converting DFA to Regular Expression

Suppose we want to find an equivalent regular expression for some DFA. Here is an algorithm to do so.

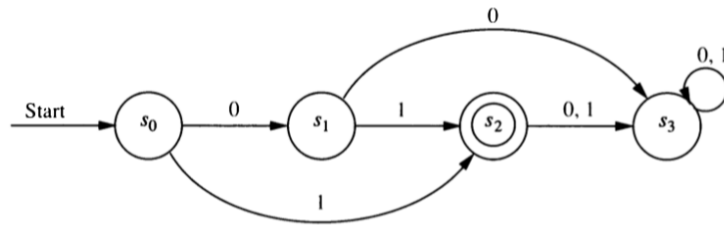
1. Modify the the original machine.
 - a. Add a **new start state** with a λ transition to the old start state. This makes the new start state **lonely**. (We can skip this step if the start state is already lonely (has in-degree 0).)
 - b. Add a **new accepting state** with a λ transition from all old accepting states. (We can skip this step if there is exactly one final state and it has out-degree 0.)
 - c. The new accepting state will be the **only accepting state**.
 - d. **Replace mutiple edges** between any pair of states A and B with one edge labeled with the union of the labels of the original edges. [Example $A \xrightarrow{0,1} B$ becomes $A \xrightarrow{0 \cup 1} B$.]
2. Pick an internal state (not the start state or the final state) to “rip out.” Let’s call that state R .
 - a. If R doesn’t have a self-loop, replace every $A \xrightarrow{r_{in}} R \xrightarrow{r_{out}} B$ with $A \xrightarrow{(r_{in}r_{out})} B$.
 - b. If R has a self-loop labeled r_{self} , replace every $A \xrightarrow{r_{in}} R \xrightarrow{r_{out}} B$ with $A \xrightarrow{(r_{in}r_{self}^*r_{out})} B$.
 - c. If this results in multiple edges between two states A and B , replace them with one edge labeled with the union of their labels.
3. Repeat step 2 until the only states left are the start state and the final state. There will be only one transition remaining, and its label will be a regular expression for the language recognized by the original DFA.

6.2.3 Examples

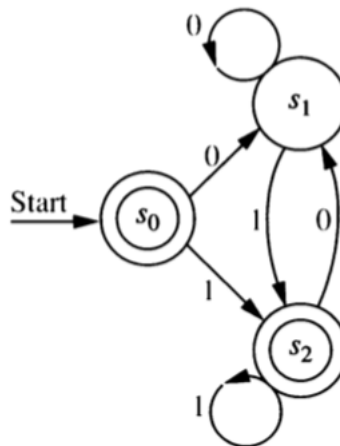
1. Systematically convert this DFA into an equivalent regular expression.



2. Systematically convert this DFA into an equivalent regular expression.



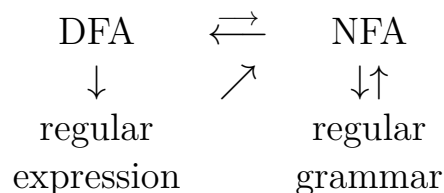
3. Systematically convert this DFA into an equivalent regular expression.



4. Would this algorithm work (possibly with slight modification) if we started with an NFA instead of a DFA? (Be sure to list any modifications that are needed.)
5. Would this algorithm work (possibly with slight modification) if we started with a GNFA instead of a DFA? (Be sure to list any modifications that are needed.)

6.3 Review problems

1. For each arrow in the diagram below, write a brief description of the algorithm that converts from one representation to another. Your description should include the most important “big idea” and also any potential “gotchas,” things you have to be careful about or might forget about.



2. Show that if A is a regular language, then $\bar{A}$ is also a regular language.
3. Show that if A and B are regular languages, then $A \cap B$ is also a regular language.
4. Convert the machines on this sheet to equivalent regular grammars.

You can find additional worked examples online at places like

- <https://courses.cs.washington.edu/courses/cse311/14sp/kleene.pdf>
- https://courses.engr.illinois.edu/cs374/sp2019/extra_notes/01_nfa_to_reg.pdf

But note that sometimes the notation isn't quite the same.

- ε is sometimes used for what we have called λ .
- $|$ is sometimes used in place of $\cup$.
- $+$ is sometimes used in place of $\cup$. (To make matters worse, $+$ is more often used for something else that we haven't seen yet; so make sure you know how $+$ is being used if you see it.)

7 A language that is not regular

Since we now have several equivalent ways to show that a language is regular, we also have several ways to show that a language is not regular.

Here is a language that is not regular: $L = \{0^n 1^n \mid n = 0, 1, 2, \dots\}$.

1. What are our options for showing the language is not regular?
2. Which do you think will be easiest?
3. See if you can give a convincing argument that L is not regular.
4. What other languages can you construct that are not regular using the same basic idea?

8 Turing Machines

The Turing machine model pre-dates electronic computers and comes from the work of Hilbert (early 1900's) and Turing and Church (1930)'s.

- Not trying to model (electronic) computers, but computation or algorithm
- Identified three essential ingredients:

1. _____
2. _____
3. _____

- implicitly there is a fourth ingredient:

4. _____

- The Turing machine is a specific formalization of these basic ingredients

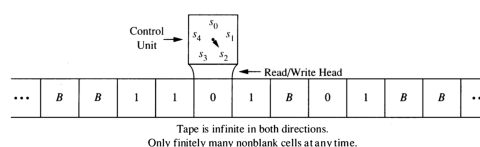
8.1 Definition of a 1-tape Turing Machine (Syntax)

A Turing machine consists of the following:

- **input alphabet**: A finite set of symbols used for inputs.
 - **tape alphabet**: Input alphabet plus and at least one additional symbol, the **blank symbol**.
 - It is important that the blank symbol is not one of the input symbols.
 - The tape alphabet may have any finite number of additional symbols beyond the blank symbol and the input alphabet.
 - Traditionally, the blank symbol is denoted $\emptyset$ or b or B or $\square$ or with an underscore.
 - **states**: A finite set.
 - **initial/start state**: One of the states to start in
 - **halting states**: Some of the states are designated as halting state. These can further be divided into accepting states and rejecting states.
 - **program**: A list of **instructions**. Each instruction has 5 parts:
 - <current tape symbol>: any tape symbol
 - <current state>: any non-halting state
 - <new tape symbol>: any tape symbol
 - <new state>: any state
 - <direction>: left, right, or stay
- * Some flavors of Turing machine do not allow stay.

8.2 Execution of a Turing Machine program (Semantics)

Imagine a machine with an infinite tape divided into cells. Each cell contains exactly one symbol from the tape alphabet. At the start of the execution, the input is written on the tape, the read/write head is located at the left most symbol of the input, and all cells that don't contain part of the input contain the blank symbol. The tape may be **1-way infinite**, in which case there is a left end to the tape, or **2-way infinite**, extending infinitely in both directions. Here is a picture of a 2-way infinite tape.



At each step in the execution of a Turing machine program, the read/write head is located at one cell of the tape. The program uses the current state and contents of the tape at that position to determine its action:

- write a symbol in the current cell (overwriting what was there before),
- move the head left or right (or stay put)
- update the state

This is repeated until one of two things happens:

- there is no instruction to execute, or
- the state is a halting state

When either of these two things happens, the machine **halts**. Otherwise it runs forever.

8.3 ASCII Representations of Turing Machines

There are several websites that provide Turing machine simulators. Here are two:

- <http://morphett.info/turing/turing.html>
 - plain output, and less flexible (initial state must always be called 0, for example), but the program encoding is terse.
- <https://turingmachinesimulator.com/>
 - nice looking and more flexible, but each instruction takes up 3 lines (current state and tape symbol on one line; new state, direction, and new symbol on the next line; then a blank line)

Each uses a slightly different ASCII representation of a Turing Machine. They are all pretty similar but differ in

- How the input alphabet, tape alphabet, start state and accepting states are described
- How they represent the blank tape symbol. (Often there is a fixed symbol for this like **b**, **B** or **_**.)
- Whether they are case sensitive, disallow certain characters, observe white space, etc.
- How left/right/stay are denoted
- The order in which the 5 parts of an instruction are given
- Which of various variations on the Turing machine theme they support (and how you tell the simulator which flavor to use).

8.3.1 Example 1 (13.5.1 in Rosen 7)

```
; in style of <http://morphett.info/turing/turing.html>
; state symbol symbol direction state
; s0 is initial state in Rosen
0 * * * s0
s0 0 0 r s0
s0 1 1 r s1
s0 _ _ r halt
s1 0 0 r s0
s1 1 1 l s2
s1 _ _ r halt
s2 1 0 r halt
```

1. For each input string below, determine the configuration (tape contents, location of read-write head, and state) of the Turing machine when it halts.
Do this by hand, then check your work using the simulator.

a. λ

b. 01

c. 0101

d. 010110

e. 001

8.3.2 Example 2

```
; in style of <http://morphett.info/turing/turing.html>
; state symbol symbol direction state
0 * * * [init]          ; [init] is "initial state"
[init] 0 0 r [10]
[init] 1 0 r [01]
[init] _ _ * halt-accept
[00]    0 0 r [10]
[00]    1 1 r [01]
[00]    _ _ * halt-accept
[01]    0 0 r [11]
[01]    1 1 r [00]
[01]    _ _ r [01]
[10]    0 0 r [00]
[10]    1 1 r [11]
[11]    0 0 r [01]
[11]    1 1 r [10]
```

2. For each input string below, determine the configuration (tape contents, location of read-write head, and state) of the Turing machine when it halts. Do this by hand, then check your work using the simulator.
 - a. λ
 - b. 0110
 - c. 01111
 - d. 01110

8.3.3 Table notation

Some authors like to represent Turing machines with tables. The goal is the same: we need to specify the action of the machine for any tape symbol and state combination. In this table the state and tape symbol are listed in the rows and columns and the action in the interior cells.

3. Convert the Turing machine in the table below

δ	a	b	blank
q0	(q0, a, R)	(q1, b, R)	(qrej, blank, L)
q1	(q1, a, R)	(qacc, b, L)	(qrej, blank, L)

into an ASCII representation for use at <http://morphett.info/turing/turing.html>.

8.4 The language recognized by a Turing Machine

We will say that a Turing machine M accepts a string x (by state) if on input x it halts in an accepting state. The language recognized by such a Turing machine is

$$L(M) = \{x \mid M \text{ accepts } x\}$$

4. If $x \notin L(M)$, what are the two ways the machine M might behave on input x ?
5. What language does the machine in Example 2 above recognize?

8.4.1 Designing a Turing Machine

For each of the following, design a Turing machine that always halts and use one of the simulators to run your program on some test strings. Explain in words what each of your states is for.

6. Design a Turing machine that recognizes $(0 \cup 1)^*1(0 \cup 1)^*$. Is every regular language recognized by a Turing machine? How do you know?
7. Design a Turing machine that recognizes $\{0^n1^n \mid n \geq 0\}$.
8. Design a Turing machine that has input alphabet $\{ (,), 0, 1, + \}$ and recognizes strings where the parentheses are properly nested.
 - Note: You don't have to do a full check that the input is a valid expression (that would be a bit more involved, go ahead and add that if you like). The intent here is just to do the parenthesis matching part.
 - Bonus: See how few states you can use.

8.5 Computing functions with a Turing machine

Turing machines can also compute functions. The output of a Turing machine is the tape contents when it halts.

9. Design a Turing machine that adds in unary. Use the input alphabet $\{1, +\}$.

In unary, a non-negative integer n is represented by $n + 1$ 1's. So 3 is represented as 1111.

Example: Our machine should convert 111+1111 into _____.

8.5.1 Neat computation

A Turing machine computes **neatly** if when it halts (a) the read/write head is on the left-most non-blank tape symbol (or any blank symbol if the tape is all blanks), and (b) there are no internal blanks (ie, all the non-blank symbols are contiguous on the tape).

10. Modify your previous machine so that it computes neatly. What is the advantage of neatly computing Turing machines?
11. Write a Turing machine that neatly computes the function $f(x) = 2x$ in unary.

9 What Can Turing Machines Do?

9.1 Turing Computable Languages

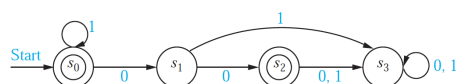
A language L is **Turing computable** (also called **decidable**) if it is recognized by some Turing machine that **always halts**, that is,

- $L = L(M)$ for some Turing machine M .
- M halts on every input. (If $x \in L$, then M halts in an accepting state on input x . If $x \notin L$, then M halts in a rejecting state on input x .)

1. Show that every regular language is Turing computable.

[Hint: What must you do to demonstrate this?]

2. Construct a Turing Machine that recognizes the same language as this DFA, using your method from the previous problem.



9.2 Variations on the Turing Machine

There are many variations on the Turing machine. Here are some examples.

- Instead of having a single tape, a Turing machine can have multiple tapes.
- Instead of being 2-way infinite, the tape(s) can have a left end. These are called 1-way infinite tapes.
- Turing machines may be required to move the head at each step (so “stay” is not an option).
- Multi-track tapes aren’t really a variation, but a handy way of thinking about a tape alphabet. (And some simulators makes this easy to do.) Since the tape alphabet may be anything, if we want to simulate writing two symbols from A , we can use a tape alphabet of $A \times A$ or even $A \cup (A \times A)$. One element of $A \times A$ represents two elements from A (one on the “top track” and one on the “bottom track”).
- Instead of having a 1-dimensional tape, a tape can be a 2-dimensional grid of cells. The read/write head can move up and down in addition to left and right.

Each of these variations is equivalent to the standard Turing machine with one 2-way infinite tape. So are many other models of computation. This has led to the so-called **Church-Turing Thesis**. Here is one statement of that thesis:

A function on the natural numbers is computable by a human being following an algorithm, ignoring resource limitations, if and only if it is computable by a Turing machine.

3. Write a 2-tape Turing machine to do parenthesis matching. It should run much faster than a 1-tape Turing machine for the same task. How will you take advantage of the second tape?
4. Give a high level description of how to convert a Turing machine with 2-way infinite tape into a Turing machine with a 1-way infinite tape.
5. Give a high level description of how to convert a 2-tape machine into a 1-tape machine.

9.3 The Halting Problem

The **Halting Problem** is a famous example of a language that is not Turing computable. By the **Church-Turing Thesis**, this would mean it is not computable by any means.

Why is it called the Halting *Problem*, you ask, instead of the Halting *Language*? This goes back to a description of languages as answers to yes/no questions. Each such problem (ie, language) can be described by an **instance** and a **question**. Here are some examples:

Euler

- Instance: A Graph G
- Question: Does G have an Euler Circuit?

3-Colorability

- Instance: A Graph G
- Question: Is G 3-colorable? (Can we color the vertices with 3 colors so that no adjacent vertices are the same color.)

SAT

- Instance: A first order logical formula φ (made with $\wedge$, $\vee$, $\neg$ and some logical variables).
- Question: Is φ satisfiable? (Can we set the variables to TRUE/FALSE in a way that makes the entire formula true?)

Implicit in each of these descriptions is some encoding scheme by which the objects mentioned in the instance are coded as bit strings. Typically, if we are only interested in whether a problem is Turing computable or not, it does not matter which coding scheme is used as long as it is systematic.¹

An entire book (Garey and Johnson 1979) of such problems (and whether or not they were NP-complete) was produced in the 1970's by Garey and Johnson and popularized this description. So you will see languages referred to as problems throughout the CS literature.

Halting Problem

- Instance: A Turing Machine M and an input string x .
- Question: Does M halt on input x ? (This is written $M(x) \downarrow$.)

¹The coding scheme can affect the efficiency of a Turing machine computation. Generally it is required that the encoding be reasonably efficient and not bloated or wasteful of space.

9.3.1 The Halting Problem is not Turing Computable (sketch)

Here is a high level sketch of the proof that the Halting Problem is not Turing computable.

1. Choose particular encoding of a Turing machines as binary strings. (This can clearly be done since our Turing machine simulators do this using ASCII strings and ASCII can be encoded in binary.)
 - Decision problems typically involve specifying an encoding. For EULER and 3-COLORABILITY, we would need to specify how we encode our graphs, for example. The details of the encoding matter very little as long as we can extract the information we need.
2. Code instances of the Halting Problem as $\langle M \rangle \# \langle x \rangle$, $\langle M \rangle$ is the encoding of M and $\langle x \rangle$ is a binary encoding of its input x .
 - The only important thing here is that we can tell where the encoding for the machine halts and the encoding for the input begins.
3. Suppose the Halting Problem is Turing Computable by a machine H . Let's use it to build a new machine K that works as follows.
 - a. on input x , create tape contents $x\#x$ and move the tape head back to the left end of the tape. We can do this by moving back and forth on our tape to copy one symbol at a time – sort of like our parenthesis matching machine.
 - b. Now run machine H with two modifications.
 - i. change all accepting states to the state LOOP and add the instruction
`LOOP * * r * ; this will cause the machine to move right forever.`
 - ii. Change all rejecting states into accepting states.

So that's our high level description of the Turing machine K . How does this help us? Let's consider $H(\langle K \rangle \# \langle K \rangle)$. (We're going to show the H gets the wrong answer.)

Exercises

6. Suppose $H(\langle K \rangle \# \langle K \rangle)$ accepts.
 - a. What does this tell us about K because of how H is defined?
 - b. What does this tell us about K because of how we defined K ?
7. Suppose $H(\langle K \rangle \# \langle K \rangle)$ rejects.
 - a. What does this tell us about K because of how H is defined?
 - b. What does this tell us about K because of how we defined K ?
8. Combine these results to explain why the Halting Problem is not Turing Computable.

9.4 Running time of a Turing machine

The running time of a Turing machine on input x is the number of steps before it halts. The conversions above typically change the running time, but machines that run in polynomial time are converted into machines that run in polynomial time (typically with a different degree polynomial).

9.4.1 P and NP

This provides a formal definition of the famous set P . P is the set of all languages that can be recognized by a Turing machine in polynomial time, that is, the number of steps the machine takes on an input of length n is bounded by $p(n)$ for some polynomial p .

A **non-deterministic Turing machine** can have multiple transitions from the same state/tape contents configuration. A non-deterministic TM accepts an input x if there is a way of choosing transitions that causes the machine to halt in a halting state. (This is similar to how non-deterministic automata were defined). NP is the set of all languages that can be recognized by a non-deterministic Turing machine in polynomial time.

The “ P vs. NP question” is whether P and NP are the same or different. To date, this question is unresolved.² Garey, M. R., and David S. Johnson. 1979. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman.

²Solving this could make you both rich and famous.